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Alkaloid Profiles and Activity in Different *Mitragyna speciosa* Strains

Luisa Boffa[a], Corrado Ghè[b], Alessandro Barge[a], Giampiero Muccioli[b] and Giancarlo Cravotto[a]*

[a]*Laboratory of Organic Chemistry;* [b]*Laboratory of Experimental Pharmacology. Dipartimento di Scienza e Tecnologia del Farmaco*, *University of Turin*, *Turin*, *Italy*

giancarlo.cravotto@unito.it

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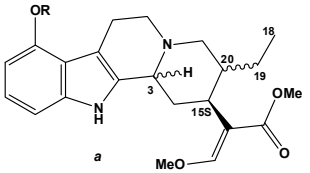
Mitragyna speciosa (K.) H. (Kratom) is a tree that possesses stimulant and opioid-like analgesic effects, and is indigenous to Southeast Asia and Indochina, where it has seen widespread use for hundreds of years. The principal pharmacologically active alkaloids in kratom leaves include mitragynine (MG), 7-hydroxymitragynine (HMG), speciociliatine (SC), speciogynine (SG) and paynantheine (P). The pharmacological effects induced and their potency can vary dramatically according to variations in the proportions of alkaloid compounds present, which are related to geographic origin, stage of maturity and ecotype. Much of the analgesic and opiate-like psychoactive effect of kratom has been associated with the MG and HMG detected in *M. speciosa* (K.). H. Five different strains of *M. speciosa* (K.) H., which present differing vein colours and geographic origin, have been studied herein; red vein strains from Thailand, Malaysia and Bali, named Red Thai, Red Malay and Red Bali, a white vein strain from Borneo (White Borneo) and a green vein strain from Malaysia (Green Malay) were included in the study. Plant leaves were extracted under magnetic stirring at room temperature in a MeOH/H_2O 1:1 mixture. Purified alkaloids were isolated in a number of organic extraction steps, from either aqueous basic or acidic phases, that culminated in precipitation (yields between 0.94 and 1.43%). These samples have been analysed using HPLC-DAD, HPLC-MS, HPLC-MS/MS and GC-MS to optimize the identification and quantification of the principal alkaloids present in the different strains. 24 alkaloids were detected in Red Bali whereas 11 compounds were found in the other varieties. Red Thai, Red Bali, Green Malay and White Borneo strains had a higher w/w percentage for MG than for P, while P was more abundant in Red Malay. The Green Malay variety (GMK) showed the highest w/w percentages for MG and total alkaloids in its extracts (59.7 and 94.9% respectively). The Green Malay variety was therefore chosen for *in vivo* pharmacological studies. The Green Malay extract has shown remarkable and significant antinociceptive and anti-inflammatory activity in mouse hot plate and carrageenan-induced paw edema tests.

Keywords: *Mitragyna speciosa* strains, Alkaloids profile, HPLC, HPLC-MS/MS analyses, Antinociceptive, Anti-inflammatory.

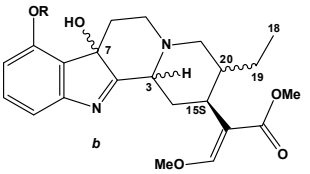
Mitragyna speciosa (K.) H. (Kratom), of the *Rubiaceae* family, is a tropical tree that is indigenous to Southeast Asia and Indochina and that grows naturally in regions including Thailand, Indonesia, Malaysia, Sumatra, Java, Bali and Borneo. Kratom has been widely used, for hundreds of years, for its stimulant and opioid-like analgesic effects [1], but it was only in the early 19th century that a Dutch botanist, named Pieter Willem Korthals, first formally documented it in Thailand [2]. The first studies focused on the isolation and identification of indole and oxindole alkaloids from *M. speciosa* (K.) H. leaves [3-7].

Botanical, phytochemical and ethnomedicinal data on the *Mitragyna* Korth genus have been collected in a recent review [8]. Brown *et al.* have reported on the presence of several alkaloids and a spectrum of other secondary metabolites, including flavonoids, polyphenolic compounds, triterpenoids, triterpenoid saponins, monoterpenes and secoirioids, in *Mitragyna* plant materials. In particular, they described 37 alkaloids that are unique to *M. speciosa* (K.) H., which was the most heavily-studied species (Figure 1). Significant amounts of pharmacologically active alkaloids, including mitragynine (MG), 7-hydroxymitragynine (HMG), speciociliatine (SC), speciogynine (SG) and paynantheine (P), can be found in kratom leaves [9].

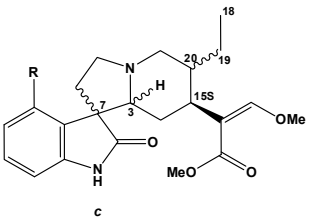
The effects of *Mitragyna* leaves can dramatically vary in effect and potency according to the differing proportions of alkaloid compounds present as well as geographic origin, stage of maturity [10,11], and ecotype [12]. The amount of MG as a fraction of total alkaloids has been documented as ranging from 66%, in plants of Thai origin [13], to 12% in plants of Malaysian origin [14].



R=OMe, 3S, 20S Mitragynine (**MG**)
R=OMe, 3S, 20R Speciogynine (**SG**)
R=OMe, 3R, 20S Speciociliatine (**SC**)
R=OMe, 3R, 20R Mitraciliatine (**MC**)
R=OMe, 3S, 20R, Δ^{18-19} Paynantheine (**P**)
R=OMe, 3R, 20R, Δ^{18-19} 3-Isopaynantheine (**IP**)
R=H, 3S, 20S Corynantheidine (**CT**)
R=H, 3R, 20S 3-Isocorynantheidine (**ICT**)



R=OMe, 3S, 7S, 20S 7-Hydroxymitragynine (**HMG**)
R=OMe, 3R, 7R, 20S 7-Hydroxyspeciociliatine (**HSC**)
R=OMe, 3R, 7R, 20R 7-Hydroxymitraciliatine (**HMC**)
R=OMe, 3R, 7R, 20R, Δ^{18-19} 7-Hydroxyisopaynantheine (**HIP**)



R=H, 3S, 7R, 20S Corynoxine B (**CB**)
R=H, 3S, 7S, 20S Corynoxine (**C**)
R=H, 3S, 7R, 20R Rhynchophylline (**RC**)
R=H, 3S, 7S, 20R Isorhynchophylline (**IRC**)
R=H, 3S, 7R, 20R, Δ^{18-19} Corynoxeine (**CE**)
R=H, 3S, 7S, 20R, Δ^{18-19} Isocorynoxeine (**ICE**)
R=OH, 3S, 7S, 20R Rotundifoline (**RT**)
R=OH, 3S, 7S, 20R, Δ^{18-19} Rotundifoleine (**RTe**)
R=OH, 3S, 7R, 20R Isorotundifoline (**IRT**)
R=OH, 3S, 7S, 20S Mitrafoline (**MF**)
R=OH, 3R, 7R, 20S Speciofoline (**SFi**)
R=OH, 3R, 7S, 20S Isospeciofoline (**ISFi**)
R=OH, 3R, 7S, 20S, Δ^{18-19} Isospeciofoleine (**ISFe**)
R=OMe, 3S, 7R, 20R, Δ^{18-19} Specionoxeine (**SNe**)
R=OMe, 3S, 7S, 20R, Δ^{18-19} Isospecionoxeine (**ISNe**)

Figure 1: Principal structures of indole (*a* and *b*) and oxindole (*c*) alkaloids of *Mitragyna spp.*

Kratom leaves produce complex dose-dependent pharmacologic effects; mild stimulant effects at lower to moderate doses (1-5 g), opioid-like effects at moderate to high doses (5-15 g), and sedative-like effects at very high doses (>15 g) [15,16]. It has been used to stave off fatigue, for the management of pain, diarrhoea and opioid withdrawal symptoms, as well as for its properties as a euphoriant.

Much of the analgesic and opiate-like psychoactive effect of kratom has been associated with the MG and HMG detected in *M. speciosa* (K.) H. [17], which are μ-opioid receptor agonists and suppress thermal and mechanical nociceptive responses (HMG's opioid agonist effect is 13 times higher than morphine's and 46 times higher than MG's) [18]. Furthermore, MG has been demonstrated to be an antagonist of both serotonergic and noradrenergic receptor systems, leading to the antinociception of both thermal and mechanical stimuli *in vivo* [19].

Methanolic extracts (50, 100, 200 mg/kg doses) and purified alkaloid kratom leaf fractions (5, 10, 20 mg/kg) have demonstrated analgesic/antinoceceptive properties when administered to mice [20]. Furthermore, possible anti-inflammatory action has been suggested as occurring through the suppression of prostaglandin E-2 production [21].

We have studied five different strains of *Mitragyna speciosa* (K.) H. that have different vein colours and geographic origin; three red vein strains from Thailand (Red Thai), Malaysia (Red Malay) and Bali (Red Bali), a white vein strain from Borneo (White Borneo) and a green vein strain from Malaysia (Green Malay).

Table 1: Extraction yields, w/w percentages of alkaloids in the extracts and concentration mg/g of plant leaves obtained by HPLC-DAD analyses.

Sample	Yield (% w/w)	% w/w Alk/EXT	Conc. mg/g Alk/PL
Red Malay	0.94	61.8	5.81
Red Bali	1.20	81.8	9.84
Red Thai	1.03	84.7	8.76
White Borneo	1.43	85.1	12.2
Green Malay	1.04	94.9	9.86

EXT = extract, PL = plant material. Alk/EXT = total alkaloids in the extract. Alk/PL = total alkaloids in the plant leaves.

Plant leaf extraction was performed in MeOH/H_2O 1:1 mixtures under magnetic stirring at room temperature for 24 h [22]. Alkaloids were purified over several steps of organic extraction, from either aqueous basic or acidic phases, culminating in precipitation, at a pH value of 9 (NH_4OH solution), filtration and drying to gave yields of between 0.94 and 1.43%. Filter residues and basic solutions were extracted with CH_2Cl_2 in order to investigate possible alkaloid losses. The various purified alkaloid samples were analyzed using HPLC-DAD (Figure 2) [23,24]. Total mean alkaloid amounts were calculated using a linear regression of the MG standard and reported as w/w mean percentage in the extract (Alk/EXT) and as mg/g concentration in plant leaves (Alk/PL) (Table 1).

Since the CH_2Cl_2 washes represented a loss of nearly 10% w/w percentage over total alkaloid extraction yields and gave very low MG and P purity, these fractions were not considered for the subsequent analyses. As Table 1 shows, the White Borneo variety gave the highest extraction yield (1.43%) and the highest amount of total alkaloids from plant leaves (12.2 mg/g PL). The Green Malay sample showed the highest w/w percentage of total alkaloids in extracts (94.9%).

Table 2: The most important peaks present in the HPLC chromatograms as w/w mean percentage.

Compound[a]	R.t.[b] (min)	*m/z*[c] $[M+H]^+$	Red Bali	Red Malay	Red Thai	Green Malay	White Borneo
ISFi+ISFe	8.52	401/399	1.26	-	-	-	-
ISFi isomer	10.3	401	0.92	-	-	-	-
ISFi isomer	11.2	401	0.94	-	-	-	-
ISFi isomer	12.0	401	0.47	-	-	-	-
IRT	13.2	401	10.1	-	tr	tr	tr
CB	14.4	385	2.15	-	tr	tr	tr
C	15.3	385	1.38	-	tr	tr	tr
CB isomer	16.3	385	0.39	-	tr	-	-
HMG isomer	17.4	415	0.81	0.2	tr	tr	-
CB isomer	18.1	385	0.29	-	tr	tr	-
CB isomer	19.4	385	0.06	-	tr	-	-
HMG isomer	20.1	415	0.71	tr	tr	tr	tr
HMG isomer	21.5	415	0.23	tr	tr	tr	tr
CT isomer	22.2	369	1.20	0.18	0.98	0.73	0.98
CT isomer	23.3	369	0.09	0.47	0.1	0.06	0.1
P	23.9	397	4.80	23.6	10.2	9.7	8.2
MG	23.9	399	37.7	4.00	44.0	59.7	48.2
CT isomer	24.4	369	0.18	0.40	0.68	0.49	0.40
SG	24.9	399	3.85	13.4	7.07	8.38	6.54
SC	25.6	399	11.0	2.33	16.6	12.0	16.0
IP	25.9	397	1.60	8.45	2.33	1.59	2.13
P isomer	26.5	397	1.59	tr	2.76	2.15	2.52
MG isomer	26.5	399	tr	7.53	tr	tr	tr
P isomer	27.4	397	-	1.18	-	0.12	0.03

[a] Compound names and formulas are shown in Figure 1. [b] Retention time in HPLC-DAD chromatograms, almost identical to HPLC-MS. [c] Mass found in ESI^+ MS chromatograms. Tr = traces.

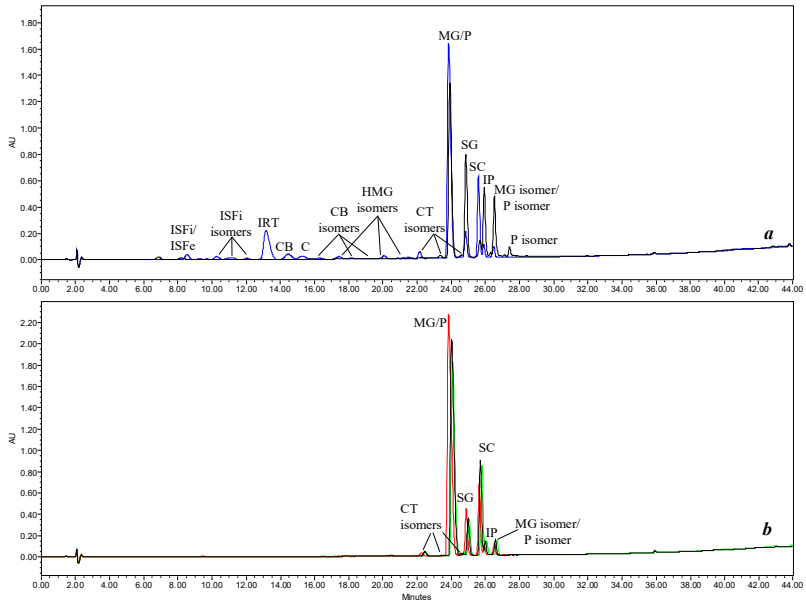


Figure 2: Comparison between HPLC-DAD chromatograms of Red Bali and Red Malay (*a*) and Red Thai/White Borneo/Green Malay (*b*).

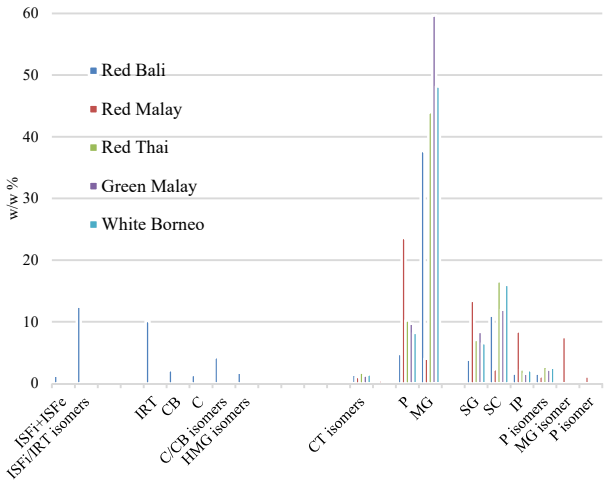


Figure 3: Comparison of the principal alkaloid compounds present in the HPLC chromatograms of the different samples.

The identification of the principal *M. speciosa* (K.) H. alkaloids present was based on a comparison of observed retention times and masses with literature values and on a combination of various complementary analytical techniques, including HPLC-MS [25], HPLC-MS/MS [26], and GC-MS [27].

As MG and P overlapped in the HPLC-DAD chromatograms, the molar ratio between them (399/397) was calculated using the single areas extracted from the HPLC-MS chromatograms, leading to separated w/w percentages of MG and P being calculated (Table 2). The Green Malay sample gave the highest w/w percentage of MG (near 59.7%). Red Thai, Red Bali and White Borneo samples displayed higher w/w percentages for MG than for P, while P was more abundant in Red Malay (23.6 vs 4%). The Green Malay extract showed the best purity in terms of total alkaloids (near 95%) (Figure 4). Purities of over 80% were obtained for Red Bali, Red Thai and White Borneo, while only around 62% was found for Red Malay.

HPLC-MS/MS chromatograms were used to identify the most important alkaloids present and to determine the masses of the non-assigned compounds.

Two of the samples analysed (Red Bali and Red Malay) were significantly different from the others, confirming the idea that different varieties and geography can cause significant changes in alkaloid distribution (Figure 3). Three isomers of corynantheidine (CT, $[M+H]^+$ at *m/z* = 369), three isomers of P ($[M+H]^+$ at *m/z* = 397), four isomers of MG ($[M+H]^+$ at *m/z* = 399) and four isomers of HMG ($[M+H]^+$ at *m/z* = 415) were found in all the samples (Table 2, Figure 3 and 4) [28].

Of these, only P and IP, at 397, and MG, SG and SC, at 399, were identified, as they were the most abundant alkaloids in all the samples. The fourth peak at 399 was principally found in Red Malay (MG isomer at 26.5 min in Table 2 and Figure 2; last chromatographic peak with *m/z* = 399 in Figure 4) and was clearly visible in GC-MS chromatograms (rt 15.86 min, Figure 5a). This peak with a P isomer (last chromatographic peak with *m/z* = 397 in Figure 4), which was overlapped present in all the other analyzed strains (Figure 3). Moreover, Red Malay presented a different compound at 397 (P isomer at 27.4 min in Table 2, last peak in Figure 2a and 3) that could only be found in traces in Green Malay and White Borneo.

Red Bali (Figure 4) contained five isomers of corynoxine B (CB) and/or hydrogenated corynoxeine (CE), with $[M+H]^+$ at *m/z* = 385, (see corynoxine (C), rhynchophylline (RC) and isorhynchophylline (IRC) in Figure 1c), and five isomers of isospeciofoline (ISFi), with $[M+H]^+$ at *m/z* = 401, (see speciofoline (SFi), rotundifoline (RT), isorotundifoline (iRT) and mitrafoline (MF) in Figure1c). Of these, CB and C, at 385, and ISFi and IRT, at 401, were indentified. A few of these compounds (especially CB, C and IRT) could only be found in traces in the other samples.

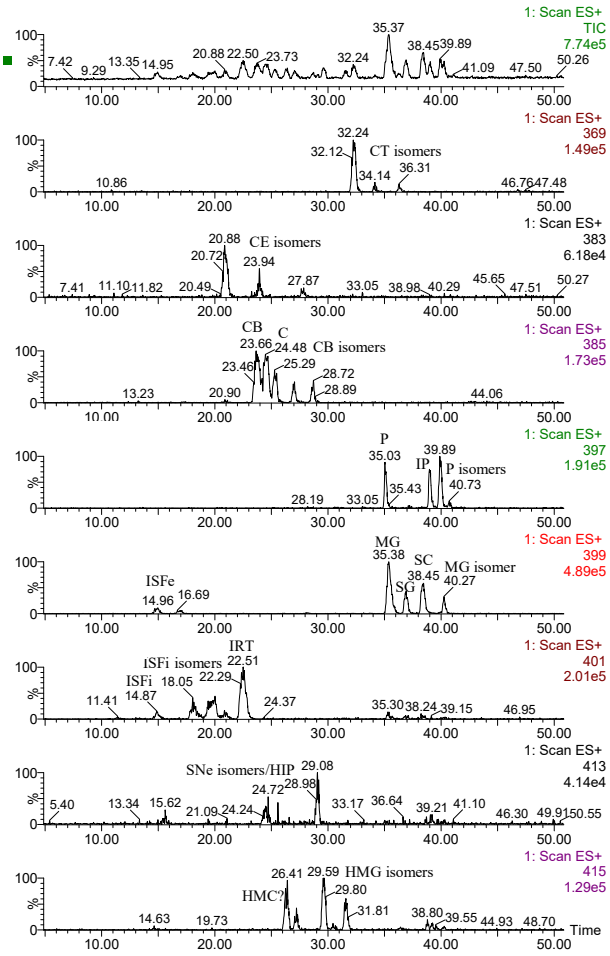


Figure 4: ESI^+ total ion (first trace) and single ion chromatograms of Red Bali (from the second to the last trace: 369, 383, 385, 397, 399, 401, 413, 415).

In the MS total ion chromatograms, some peaks were found with $[M+H]^+$ at *m/z* = 383 and 413. These correspond to CE or isocorynoxeine (ICE) and specionoxeine (SNe), isospecionoxeine (ISNe), 7-hydroxyisopaynantheine (HIP) or isopaynantheine-*N*(4)-oxide (Figure 1c and 4). Some of these compounds were first described by Cao *et al.* in 2013 [29].

Several articles in the literature have described a GC method for the analysis of both underivatized and TMS derivatives of MG and other indole alkaloids [30-32]. In particular, Wang *et al.* characterized two sets of indole alkaloid diastereoisomers (P and IP and MG, SG and SC) and two oxindole alkaloid diastereoisomers, C and CB in 2014 [33]. They demonstrated that the GC method had two major problems, the high temperature required to elute the alkaloids and inadequate resolution to distinguish MG and SC

(diastereoisomers differing only in the orientation of an interior hydrogen atom), and therefore proposed that only GC-MS be used for the separation and identification of these compounds in kratom. In our work, we have applied the method used by Wang *et al.* and identified almost all of the compounds present in our samples. CB (rt 12.62, $M^{\cdot+}$ *m/z* = 384) and C (rt 12.769, $M^{\cdot+}$ *m/z*=384) were only detectable in the Red Bali strain (Figure 5b), while IP (rt 15.67, $M^{\cdot+}$ 396) was found in the Red Malay strain (only traces in Red Bali) (Figure 5a). MG/SC (rt 17.88, $M^{\cdot+}$ *m/z* = 398), P (rt 18.33, $M^{\cdot+}$ *m/z* = 396) and SG (rt 18.76, $M^{\cdot+}$ *m/z* = 398) were found in all the analysed samples.

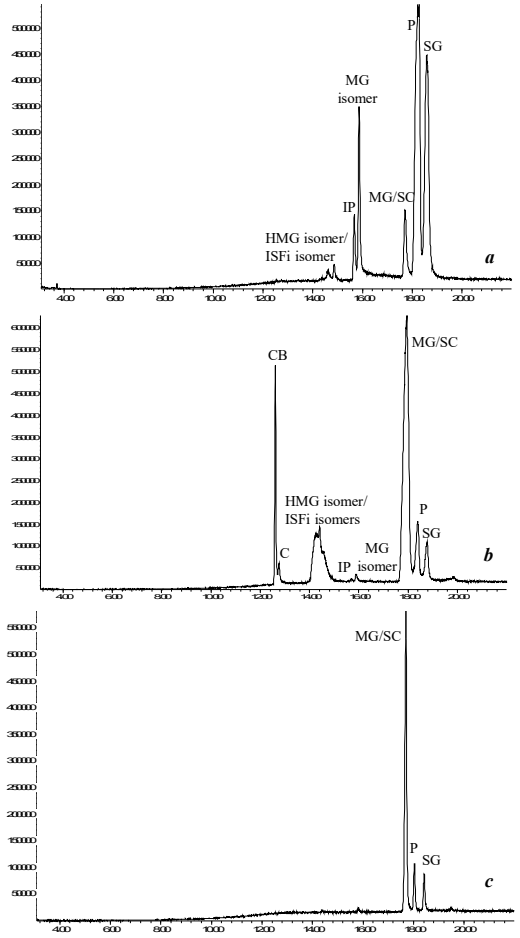


Figure 5: GC-MS chromatograms of Red Malay (a), Red Bali (b) and Red Thai/White Borneo/Green Malay (c) extracts.

HMG isomers (rt 14.65, $M^{\cdot+}$ *m/z* = 414) can be difficult to identify in Red Bali and Red Malay strains because they overlap with other peaks and are present in low amounts. New alkaloids were found with $M^{\cdot+}$ 400 (broaden peak, rt ranging from 14.2 to 14.6) and $M^{\cdot+}$ *m/z* = 398 (rt 15.86) in the Red Bali and Red Malay samples. GC-MS libraries (NIST, Wiley) gave IRT and SFi as possible structures for $M^{\cdot+}$ *m/z* = 400, while the structures SC or MC were proposed for $M^{\cdot+}$ *m/z* = 398.

As can be seen in Figure 5a and b, GC-MS analyses also highlight that the distributions of MG and P in Red Malay are different to those of the other samples. However, HPLC is to be preferred for the analysis of *M. speciosa* (K.) H. extracts since it provides better separation and sensitivity. As shown in Table 2, the Green Malay strain was the richest in MG and was therefore chosen for pharmacological studies.

To investigate whether the Green Malay extract possesses any of the typical pharmacological activities (i.e. analgesic and anti-inflammatory effects) that have been widely described, in the literature, as being displayed by *M. speciosa* (K.) H. [8], the bio-activity of its extract was evaluated in two classic animal tests; the hot-plate pain test, where opioid neurotransmission is involved, [34] and carrageenan-induced paw edema, an acute inflammation mainly mediated by prostaglandin synthesis [35]. The Green Malay extract showed a significant anti-nociceptive effect within 90 min of treatment. The effect peaked between 15 and 30 min and decreased after 60 and 90 min (Figure 6A). The Green Malay extract also displayed an evident anti-inflammatory activity which reduced, within 6 h, the development of the paw edema that was induced by a subplantar carrageenan injection (Figure 6B).

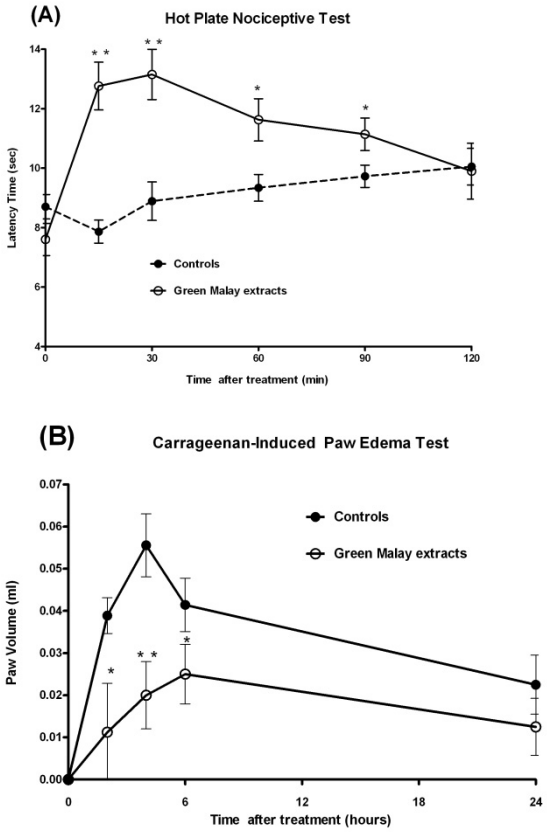


Figure 6: The effect of the Green Malay extract on mouse hot-plate nociceptive (A) and carrageenan-induced acute inflammatory paw edema tests (B). Mice were treated orally with 10 ml/kg of vehicle (Controls) or 10 ml/kg of the Green Malay extract containing 20 mg/kg MG. Animals were analyzed at indicated times. Values are mean ± S.E. of six animals per group. **P*< 0.05, ***P*< 0.01 vs controls.

Experimental

General: The MG standard was purchased from ChromaDex (Irvine, CA, USA). All solvents (petroleum ether - PE, ethanol - EtOH, dichloromethane - CH_2Cl_2) and reagents (ammonium hydroxide – NH_4OH, chloridric acid – HCl) were purchased from Sigma Aldrich at analytical grade level (Milan, Italy). Methanol (MeOH), acetonitrile (CH_3CN) and trifluoroacetic acid (TFA), used for HPLC analyses, were purchased from Sigma-Aldrich at HPLC grade. Water was purified using a Milli-Q Reference A+ System (Merck Millipore, Bedford, USA).

Plant material: The leaves of three red vein strains of *M. speciosa* (K.) H., from Thailand (Red Thai), Malaysia (Red Malay) and Bali (Red Bali), a white vein strain from Borneo (White Borneo) and a green vein strain from Malaysia (Green Malay) were all provided by Herba Invest S.R.O. (Bratislava, Slovakia).

Extraction and purification: The plant material (50 g) was placed into a round bottomed flask (1 L), 500 ml of a MeOH/H_2O 1:1 mixture were added and the suspension was kept under magnetic stirring at room temperature for 24 h.

The suspension was filtered on paper in a Büchner funnel and MeOH was evaporated under vacuum.

A 0.1 M NH_4OH solution was added to the residual suspension to give a pH value of 9. The resulting suspension was extracted with CH_2Cl_2, which was evaporated to give the crude extract.
The extract was dissolved in HCl 0.1 N (pH = 3) and washed with petroleum ether (3 times).

A 0.1 M NH_4OH solution was added to the acidic solution to give a pH value of 9. The alkaloids precipitated. The precipitate was filtered on paper in a sintered glass filter, dried and weighed, while the residue on the filter and the basic solution were extracted with CH_2Cl_2.

HPLC-DAD analyses: MG standard and alkaloid extract solutions were prepared via dissolution in ethanol and 1 to 10 dilution in a mixture of H_2O/CH_3CN 2:1 (TFA 0.1%). The solutions were filtered through a 0.45 µm syringe filter and analyzed by HPLC on an XTerra MS C8 column (4.6 x 150 mm, 5 µm, Waters). The PDA detector was. The analyses were carried out on a Waters 1525 Binary HPLC pump equipped with 2998 PDA, using a flow rate of 1 mL/min and an injected volume of 20 µL. The quantification of the MG standard and all the alkaloids in the extracts was performed at 222 nm. 0.1% TFA aqueous solution (A), and acetonitrile, with 0.1% TFA (B), were used as the mobile phases accordingly to the following gradient profile: (time, B%) 0.01, 20; 7.5, 20; 15, 30; 26, 60; 39.5, 100; 44, 100.

HPLC/MS analyses: HPLC/MS analyses were carried out on a Waters Fraction Link autopurification system equipped with Waters 2996 PDA, Waters 2525 binary pump and Waters Micromass ZQ detector operating in ESI+ mode. Chromatographic separations were carried out using the same method described for HPLC-DAD analyses.

HPLC-MS-MS analyses: HPLC-MS-MS analyses were carried out on a UPLC Acquity Waters system equipped with Binary Solvent Manager, Sample Manager, Column Manager, PDA and Micromass Quattro microTM operating in ESI+ mode. 5 µL of extract solution was loaded on an XTerra MS C8 column (4.6 x 150 mm, 5 µm, Waters) and eluted with 0.1% TFA in water (A), and 0.1% TFA in acetonitrile, (B) (gradient: time, B%: 0.01, 20; 15, 20; 30, 30; 52, 60; 68, 100; 80, 100) using a flow rate of 0.5 mL/min.

GC-MS analyses: GC-MS qualitative analyses were performed in an Agilent Technologies 6850 Network GC System equipped with a 5973 Network Mass Selective Detector and a 7683B Automatic Sampler, using a capillary column (HP-5MS 5% Phenyl Methyl Siloxane, length 30 +10 m; i.d. 0.25 mm; film thickness 0.25 µm). A flow rate of 1 mL/min, an injection volume of 1 µL and a split ratio of 25:1. The column temperature ramp was programmed from 220°C (2 min) to 300°C at 8°C/min and then held at 300°C for 10 min. The scanned mass range was 35–550 Da.

Pharmacological tests: Male albino Balb/C mice were used. Animals were fasted for 24 h before treatment. The nociceptive test was performed using a Hot-plate instrument (mod. 35100, Ugo Basile, Italy), set at 55 ± 0.2°C as described by Matsumoto *et. al* [18]. The acute inflammatory test was performed using a subplantar carrageenan-induced paw edema, as previous described [36]. Paw edema was measure using a plethysmometer (mod. 7140, Ugo Basile, Italy). Statistical analyses were performed using one-way ANOVA followed by the Newman Keulse test.

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